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SLIATE

SRI LANKA INSTITUTE OF ADVANCED TECHNOLOGICAL EDUCATION

(Established in the Ministry of Higher Education, vide in Act No. 29 of 1995)

Higher National Diploma in Quantity Surveying
2nd Year, 1st Semester Examination- 2023
QS 21013 – Measurement II

Instructions for Candidates:
Answer 03 questions only
Answering Question 01 is Compulsory

No. of questions : 05
No. of pages : 07
Time : **03 hours**

- Clean unmarked copy of standard Method of Measurement of Building Works (SLS 573) is permitted to use in the exam.
- You may use the below given equation when calculating reinforcement requirement Unit weight of steel (kg/m) = (Diameter of steel in mm) ^{2}/162.162
- All drawings are not to scale.
- Assume the width of sun shed as 1200 mm.
- Assume the unit weight of Gusset plate as 48.77 kg/m2 and unit weight of truss frame as 4.43 kg/m.
- Assume the width of reveals and skirting height as 75mm.
- Assume the clear cover as 50mm for all elements.

Question 01

The following figures show a floor plan of the 1st floor in a two storied building. Take of quantities for the following items with reference to the floor plans, cross sections and foundation details given below ***(Figures 2,3,4,5 & 6).***

Use of neat sketches, clear calculations and sensible assumptions following the relevant SMM rules when preparing the TDS sheets is compulsory).

Take-off quantities for the following items.

I. In-situ Concrete Works at Column foundation (05 Marks)
II. Formwork for Column foundation (05 Marks)
III. Reinforcement for Column foundation (08 Marks)
IV. Random Rubble work at Wall foundations (06 Marks)
V. In-situ Concrete for plinth beam (04 Marks)
VI. Formwork for plinth beam (04 Marks)
VII. Reinforcement for plinth beam (08 Marks)

(Total 40 Marks)

Question 02

Take off quantities referring to the same figures as used in Question 01 ***(Figures 2,3,4,5 & 6)***.

a. Prepare TDS for the Masonry Work items stated below

i. Block work for external walls (03 Marks)

ii. Block work for internal walls (03 Marks)

b. Prepare TDS for wall finishes

i. Plastering work for external walls (03 Marks)

ii. Painting work for beams and columns and external walls (03 Marks)

iii. Plastering work for doors and window reveals (03 Marks)

iv. Painting work for doors and window reveals (03 Marks)

b. Prepare a BOQ for items measured in Question 01 and part (a) and (b) of Question 02 (12 Marks)

(Total 30 Marks)

Question 03

Internal finishes hold a greater cost concern in project estimations. Internal finishes such as til and granite finishes, ceiling finishes are vital in creating the final aesthetic and functiona surfaces in key areas of the house construction project.

a. Takeoff measurements for floor and ceiling finishes ***(Figures 2,3,4,5 & 6).***

I. Tile work for Bathroom walls (04 Marks)

II. Tile work for floor area (04 Marks)

III. Granite finishing to kitchen counter work top (04 Marks)

IV. Skirting Tile Work (04 Marks)

V. Ceiling Finishes bed room and master bedroom (04 Marks)

b. Describe the rules and principles for measuring tiling works according to the Standar Method of Measurement of Building Works (SLS 573). Include how measurements ar taken for wall and floor tiles and how allowances for openings and decorative patterns ar addressed. (06 Marks

c. Briefly explain 02 coverage rules and 01 supplementary information related to interna finishes of a building. (04 Marks)

(Total 30 Marks)

Question 04

a. Takeoff quantities for structural Steel works at a roof truss ***(Figure 6).***

   i. Design and Framing fabrication of Truss work (Assume total no of truss as 3) (03 Marks)

   ii. Permanent erection of truss work (03 Marks)

   iii. Purlins (Assume the number of purlin per slope as 2) (03 Marks)

   iv. Special Bolts (Assume the number of bolts per each connection as 3) (03 Marks)

   v. Gusset plates (03 Marks)

b. Prepare a BOQ for the measurements taken for part (a) in Question 04. (08 Marks)

c. Compare the Provisional Sum (PS) and Prime cost Sum (PC Sum) in a construction project. (04 Marks)

d. Explain the importance of Preliminaries to a Construction Project (03 Marks)

(Total 30 Marks)

Question 05

a. Take- off quantities for following work items in a staircase drawing ***(Figure 01).***
(Assume the number of stairs as 22, width of a stair is as 1200mm and thick of slope as 200mm)

   I. Reinforcement at stair stringer/ slope (05 Marks)

   II. Formwork for steps (03 Marks)

   III. Formwork for stair stringer/ slope (03 Marks)

   IV. Concrete for stairs (04 Marks)

   V. Concrete for stair stringer/ slope (04 Marks)

b. Briefly explain preambles in a BOQ (03 Marks)

c. What are the sections to be used when measuring the followings?

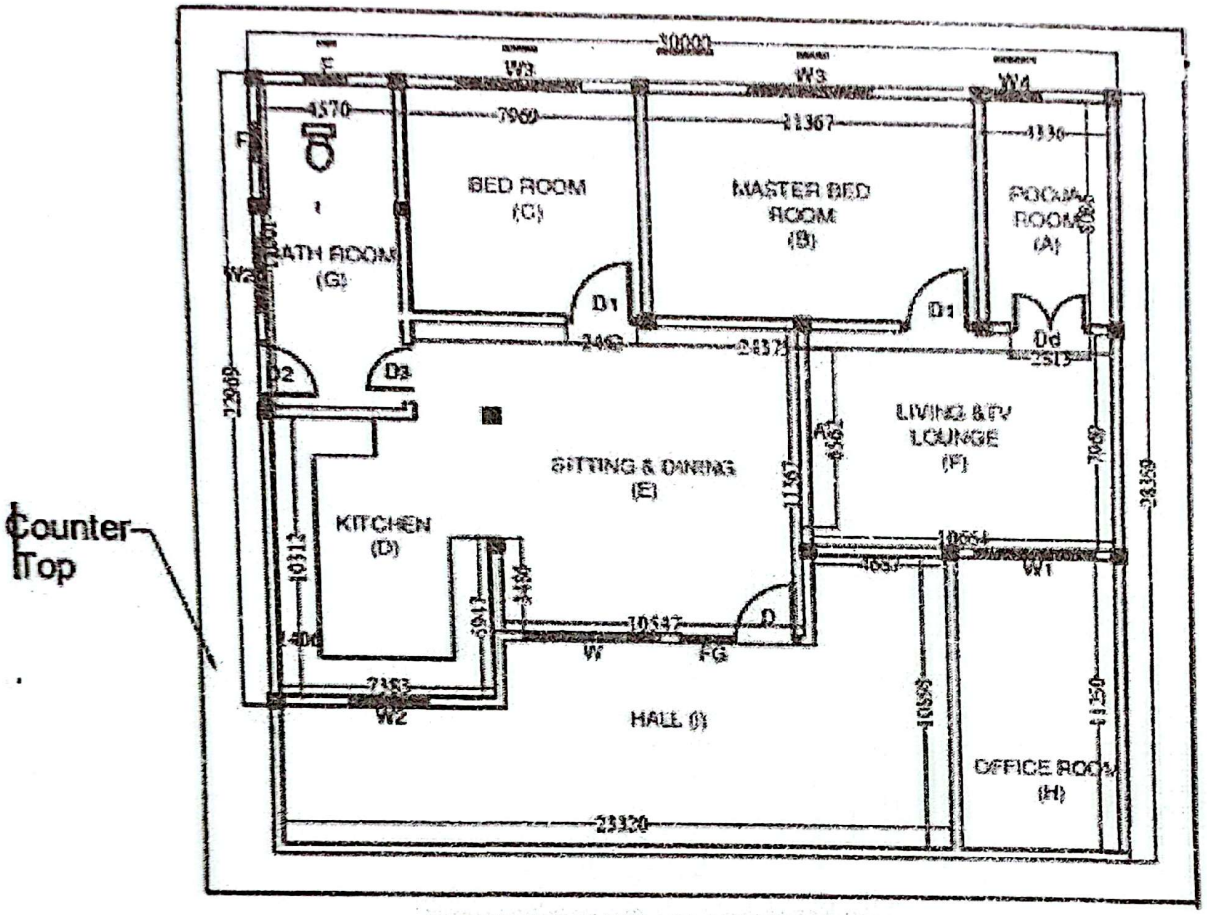
   I. Proprietary cladding

   II. Demolition works

   III. Alteration works

   IV. Underpinning works (01 X 4 = 04 Marks)

d. State the guidelines to be followed when taking off quantities for Works to existing buildings. (04 Marks)



FIRST FLOOR PLAN OF TWO STORIED BUILDING

Figure 2: 1st Floor Plan

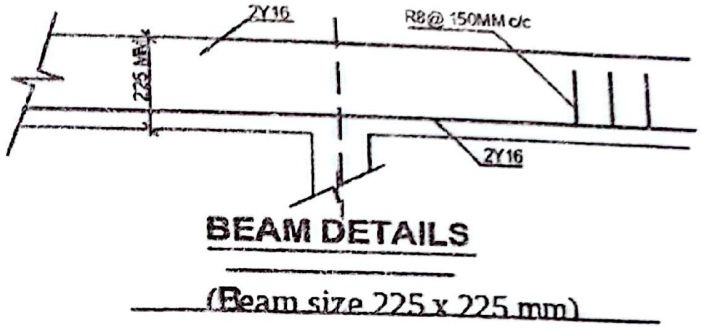


Figure 3: Beam Details

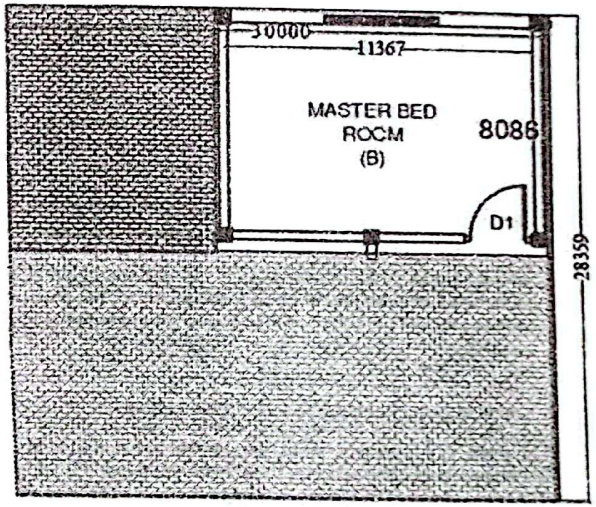


Figure 4: 2nd Floor Plan

Assume the height of each floor as 3300mm.

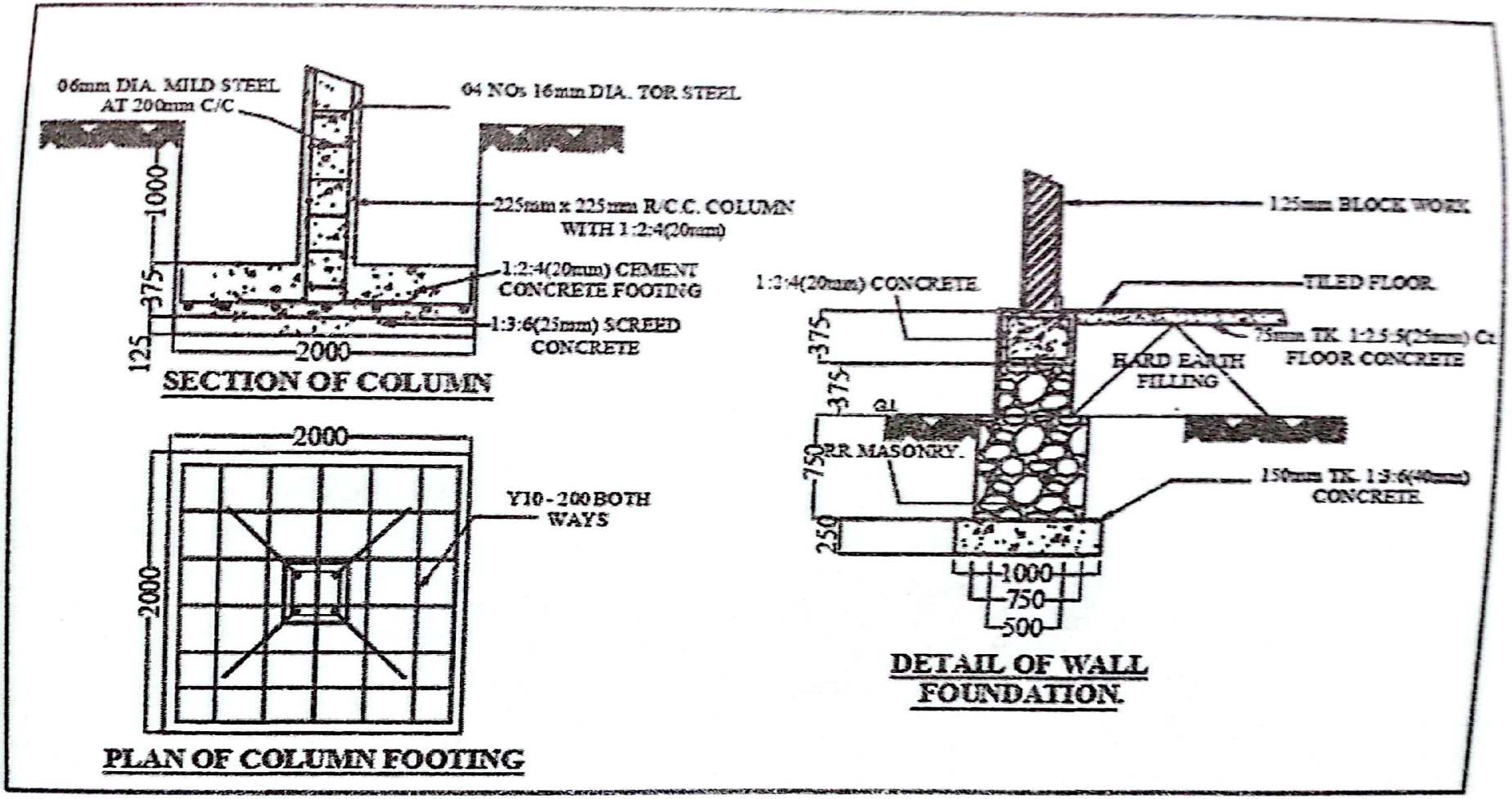
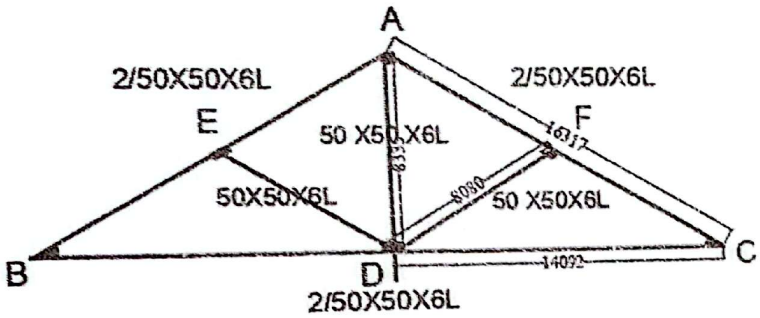


Figure 5: Details of the Foundations



CROSS SECTION OF STEEL ROOF.

SCHEDULE OF WALL OPENINGS OF PROPOSED HOUSE

TYPE	SIZE	DESCRIPTION	NOS
Dd	1.20 X 2.30	DOUBLE PANELED TOP FANLIGHT	02
D	1.20 X 2.25	PANELED TOP FANLIGHT	03
D1	1.05 X 2.25	PANELED TOP FANLIGHT	05
D2	0.90 X 2.00	PANELED DOOR	02
D3	0.80 X 1.95	PANELED MARINE DOOR	04
W	2.00 X 1.65	GLAZED WITH TOP LOUVERS	02
W1	1.65 X 1.65	GLAZED WITH TOP LOUVERS	02
W2	1.20 X 1.50	GLAZED WITH TOP LOUVERS	04
W3	1.65 X 1.50	GLAZED WINDOW	04
W4	0.75 X 1.50	GLAZED WINDOW	02
FG	0.85 X 2.25	FRENCH AND FIXED GLASS	02
F	0.60 X 0.45	GLAZED FAN LIGHT	04
A	8 X 2.40	ARCH AS PER DIMENSIONS	01

Size of Gusset Plates

GP A- 225x150
GP- B,C- 225X160
GP- E,F- 200X100
GP- B-150X150

PROPOSED HOUSE PLAN FOR MR & MRS KRISHNA, NO 12, WAIMAN ROAD, NALLUR, JAFFNA	SIG. OF APPLICANT	SIG. OF QUALIFIED PERSON	SIG. OF CHARTED EBGINEER .....

Figure 6: Truss Details, Doors and Windows Schedule

(Total 30 Marks)

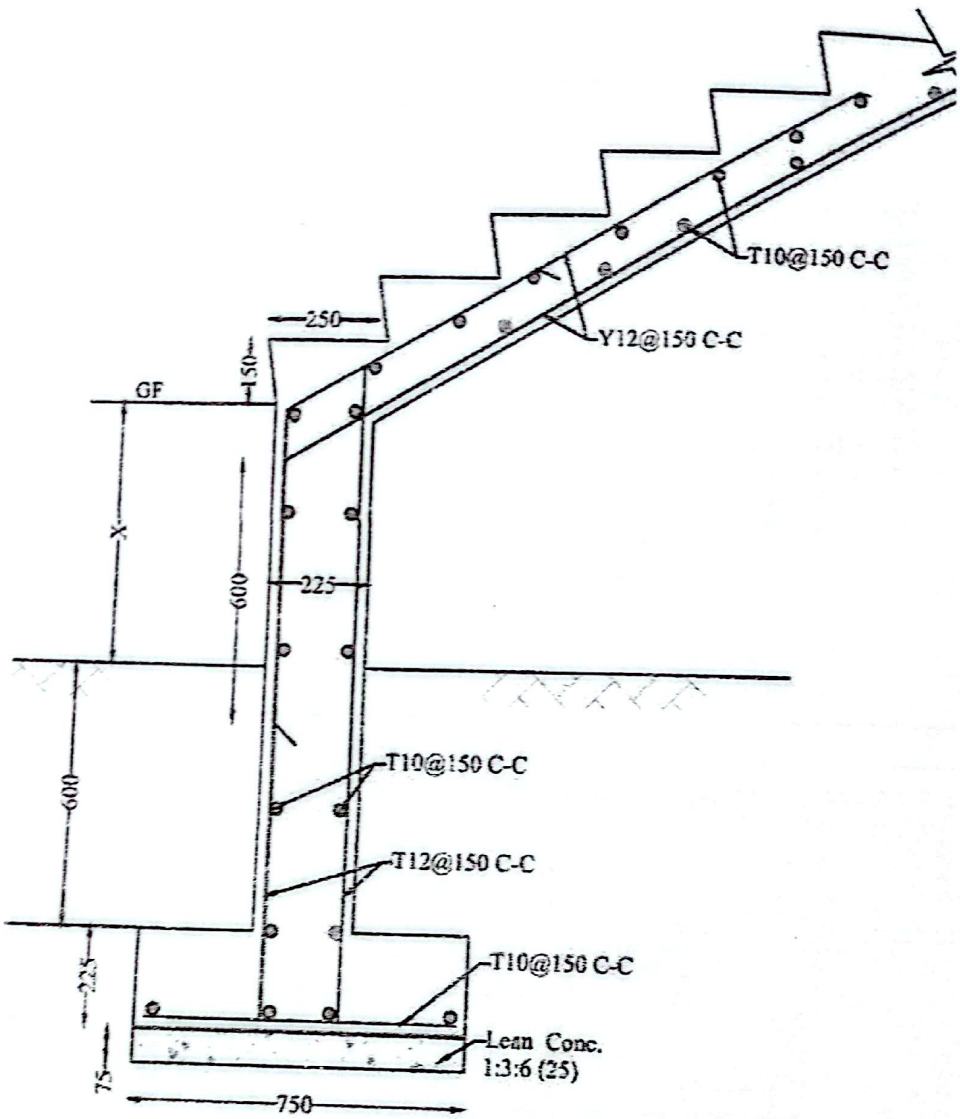


Figure 1: Cross Section of Staircase